

Instructions:

1. This question paper contains 50 questions. All questions are compulsory.
2. The duration of the examination is 90 minutes.
3. Each question carries 2 marks. There are no negative marks for wrong answers.
4. Use only blue or black ball-point pen to mark your answers on the OMR Answer Sheet.
5. The use of calculators, mobile phones, smart watches, log tables, or any electronic gadgets is strictly prohibited.

Science

1. A scientist discovers a microscopic entity in a thermal spring. He observes that it does not breathe, does not take in nutrients, and shows no signs of movement. However, when it enters a specific bacteria's body, it begins to multiply rapidly. Based on the characteristics of living organisms, how should this entity be classified?
(A) It is a living organism because it can reproduce.
(B) It is a non-living object because it cannot move or breathe on its own.
(C) It lies on the borderline of living and non-living things.
(D) It is a plant because it lives in water.
2. A carpenter needs to cut a wooden plank. He places a broken ruler on the plank such that one end of the plank is at the 3.5 cm mark and the other end is at the 18.2 cm mark. What is the actual length of the wooden plank?
(A) 18.2 cm (B) 15.7 cm
(C) 14.7 cm (D) 21.7 cm
3. Classification and Properties of Materials
Four materials—A, B, C, and D—are described below:

Material	Property 1	Property 2	Use Case
A	Can be stretched without breaking	Poor conductor of electricity	Used in ropes and bags
B	Hard, brittle	High melting point	Used in cutting tools
C	Conducts electricity well	Malleable and ductile	Used in electrical wires
D	Soft, malleable	Poor conductor	Used in jewellery

Which correct combination of material type and examples is most plausible?

- (A) A → Synthetic polymer (nylon), B → Ceramic (glass), C → Metal (copper), D → Non-metal (sulfur)
 (B) A → Natural polymer (cotton), B → Allotrope(diamond), C → Metal (aluminum), D → Metal (gold)
 (C) A → Synthetic polymer (polyester), B → Allotrope (diamond), C → Metal (copper), D → Metal (gold)
 (D) A → Composite material (fiberglass), B → Alloy (steel), C → Metal (silver), D → Non-metal (wax)

4. A scientist is studying a mysterious organism found in a deep-sea hydrothermal vent. The organism lacks a stomach but has a specialized outer membrane that absorbs dissolved organic molecules. It also doubles its population every 20 minutes in a lab setting. **Which two characteristics of life are being directly described here?**

(A) Respiration and Excretion
(B) Nutrition and Reproduction
(C) Response to Stimuli and Movement
(D) Adaptation and Respiration

5. Arrange the steps of **"Correct Measurement"** in order:

1. Calculate the difference between end and start readings.
2. Place the scale along the length of the object.
3. Note the reading using vertically above the mark.
4. Check for zero error.

(A) 1 -> 2 -> 3 -> 4 (B) 2 -> 1 -> 4 -> 3
(C) 4 -> 2 -> 3 -> 1 (D) 1 -> 3 -> 2 -> 4

6. A student has a mixture of ethanol, water, and sand. The student wants to separate these components and obtain them in pure form. **Which of the following sequences is most appropriate?**

(A) Filter the mixture → evaporate water → distill ethanol → collect sand.
(B) Distill ethanol → filter the residue → evaporate water → collect sand.
(C) Filter the mixture → distill ethanol → evaporate water → collect sand.
(D) Evaporate water → distill ethanol → filter → collect sand.

7. A specific ecosystem contains a **"Producer,"** a **"Herbivore,"** and a

"Carnivore." If a toxin that does not break down is introduced into the plants, it accumulates more in the bodies of the animals higher up the food chain. This is a concept related to how living things interact. What is the primary characteristic of the **"Producer"** in this chain?

(A) It is a heterotroph that consumes waste.
(B) It is an autotroph that converts solar energy into chemical energy.
(C) It is a decomposer that breaks down dead matter.
(D) It is a scavenger that cleans the environment.

8. Identify the "Odd One Out" based on the type of motion:

(i) Movement of a pendulum
(ii) Movement of a swing
(iii) Heartbeat of a healthy person
(iv) Blinking of eyes

(A) Pendulum (B) Swing
(C) Heartbeat (D) Blinking of eyes

9. A chemistry student sets up an experiment to burn magnesium ribbon in three different conditions: (1) pure oxygen, (2) normal air, and (3) a nitrogen-rich environment with only 10% oxygen. Observations showed the brightest flame and fastest reaction in pure oxygen, a moderate flame in air, and a slow or nearly no reaction in nitrogen-rich air.

Which principle most logically explains the difference in the reactions?

(A) Combustion requires oxygen because it acts as a fuel, so higher oxygen concentration accelerates burning.

(B) Oxygen is an oxidizing agent, so higher oxygen concentration increases the rate of oxidation, while nitrogen is largely inert and does not support combustion.

(C) Magnesium reacts only with oxygen molecules, so other gases in the air inhibit the reaction.

(D) Oxygen concentration does not affect combustion; the difference is due to varying temperatures in the setups.

10. A biologist compares two regions. Region X has 1,000 organisms, but they all belong to only 2 species. Region Y has 1,000 organisms belonging to 150 different species. Which of the following statements is scientifically accurate?

(A) Region X has higher biodiversity because the population of each species is large.

(B) Region Y has higher biodiversity because there is a greater variety of life forms.

(C) Both regions have equal biodiversity because the total number of individuals is 1,000.

(D) Diversity cannot be compared without knowing if the organisms are plants or animals.

11. A toy train moves on a circular track of radius 7 m. The train completes 10 rounds. (Take $\pi = \frac{22}{7}$). Calculate the total distance covered.

(A) 44 m (B) 154 m
(C) 440 m (D) 220 m

12. An engineer must choose a material for four different applications:

1. A water bottle that is light and shatterproof

2. A kitchen knife blade that must remain sharp

3. Electrical wiring in a building

4. Decorative jewelry that does not tarnish

Which of the following sets of materials is most suitable?

(A) 1 → Glass, 2 → Aluminum, 3 → Copper, 4 → Silver

(B) 1 → Plastic, 2 → Steel, 3 → Copper, 4 → Gold

(C) 1 → Metal alloy, 2 → Ceramic, 3 → Aluminum, 4 → Platinum

(D) 1 → Wood, 2 → Diamond, 3 → Iron, 4 → Copper

13. Consider the following biological "Excretory Products":

1. Oxygen during the day.
2. Carbon Dioxide during the night.
3. Water vapor through leaves.
4. Dry, fallen leaves in autumn.

Which organism is likely responsible for all these types of excretion?

(A) An Earthworm (B) A Neem Tree
(C) A Human Being (D) A Mushroom

14. Read the following statements about the "Kink" in a clinical thermometer:

Statement I: The kink allows the mercury to expand easily when heated.

Statement II: The kink prevents the mercury from falling back into the bulb on its own.

Statement III: The kink is present in both laboratory and clinical thermometers.

Choose the correct evaluation:

(A) Only Statement II is true.
(B) Statements I and II are true.
(C) Statements II and III are true.
(D) All statements are false.

15. Which property is typical of ceramics?

- (A) Flexible and stretchable
- (B) Good conductor of electricity
- (C) Hard and brittle
- (D) Soft and malleable

16. Consider the following observation: An Amoeba (single-celled organism) is placed in a drop of water. When a tiny grain of salt is added to the left side, the Amoeba slowly moves to the right. This is a direct example of which two life characteristics?

- (A) Reproduction and excretion.
- (B) Nutrition and growth
- (C) Respiration and diversity.
- (D) Response to stimuli and locomotion.

17. Geeta wants to magnetize an iron needle using the "Stroking Method." She places the needle on a table and strokes it with the North Pole of a bar magnet from end A to end B. She lifts the magnet, brings it back to A, and repeats the process. What will be the polarity formed at end B of the needle?

- (A) North Pole
- (B) South Pole
- (C) No Pole is formed
- (D) Magnetic Equator

18. A city suffers from smog during winter. Residents complain of breathlessness and fatigue. Scientists measure oxygen levels and find they are slightly lower than usual, while carbon monoxide and particulate matter are high.

From a critical thinking perspective, what is the most plausible explanation for these health issues?

- (A) Oxygen is chemically destroyed by smog, leading to lower oxygen concentration.

(B) Carbon monoxide binds to haemoglobin more effectively than oxygen, reducing the blood's oxygen-carrying capacity, which explains fatigue even though oxygen levels in the air are nearly normal.

(C) Smog increases oxygen levels, but people cannot absorb it because their lungs are blocked.

(D) Air pressure decreases due to pollution, which reduces oxygen availability for everyone equally.

19. **Assertion:** We should include a variety of food items in our daily meals rather than just one type of nutrient.

Reason: No single food item contains all the essential nutrients required by the body in the correct proportions.

(A) Both Assertion and Reason are true, and Reason is the correct explanation of Assertion.

(B) Both Assertion and Reason are true, but Reason is NOT the correct explanation of Assertion.

(C) Assertion is true, but Reason is false.

(D) Assertion is false, but Reason is true.

20. Match the magnet shapes in Column I with their typical descriptions or common uses in Column II.

Column I	Column II
(P) U-shaped Magnet	(i) Used in magnetic compasses
(Q) Magnetic Needle	(ii) Stronger than bar magnets, poles closer
(R) Disc Magnet	(iii) Ends look like circular planes
(S) Bar Magnet	(iv) Classic rectangular lab magnet

(A) P-ii, Q-i, R-iii, S-iv

(B) P-iv, Q-iii, R-ii, S-i

(C) P-ii, Q-iv, R-i, S-iii

(D) P-i, Q-ii, R-iv, S-iii

21. A chemist wants to separate a mixture of three substances: oil, water, and chalk powder. The chalk is insoluble in both water and oil. **The chemist proposes the following steps:**

1. Use a separating funnel to remove water.
2. Filter the residue to remove chalk.
3. Decant oil.

Which statement is correct?

- (A) Step 1 should be done after step 2.
- (B) Chalk can be removed by filtration (or sedimentation/decantation) before separating oil from water.
- (C) Oil and water can be separated using a funnel because they are immiscible.
- (D) The method will not give pure substances.

22. Three **cars (Car 1, Car 2, Car 3)** have bar magnets fixed on their roofs.

- Car 1 has its North pole facing forward.
- Car 2 is in front of Car 1 and has its South pole facing backward (towards Car 1).
- Car 3 is behind Car 1 and has its North pole facing forward (towards Car 1).

What will happen to Car 1?

- (A) Attracted by Car 2 and repelled by Car 3 (Moves forward).
- (B) Repelled by Car 2 and attracted by Car 3 (Moves backward).
- (C) Attracted by both Car 2 and Car 3 (Stays in middle).
- (D) Repelled by both.

23. A student wants to obtain solid salt from seawater. **Which method is most suitable?**

- (A) Filtration
- (B) Evaporation
- (C) Sublimation
- (D) Distillation

24. Which two planets in our solar system do NOT have any natural satellites ?

- (A) Earth and Mars
- (B) Mercury and Venus
- (C) Jupiter and Saturn
- (D) Uranus and Neptune

25. Which statement about chromatography is true?

- (A) It separates components based on solubility and polarity.
- (B) It separates substances based on boiling point.
- (C) It is only used for solids.
- (D) It can separate only two components at a time.

26. If the distance between the Sun and Earth (1 AU) is approximately 150 million km, and Jupiter is 5.2 AU from the Sun, what is the distance of Jupiter from the Sun in kilometers?

- (A) 300 million km
- (B) 650 million km
- (C) 780 million km
- (D) 1000 million km

27. **Assertion:** A dry seed in a packet is considered a living thing.

Reason: Seeds are in a state of "dormancy" where life processes are very slow but active.

- (A) Both Assertion and Reason are true, and Reason is the correct explanation of Assertion.
- (B) Both Assertion and Reason are true, but Reason is NOT the correct explanation of Assertion.
- (C) Assertion is true, but Reason is false.
- (D) Assertion is false, but Reason is true.

28. Air is a mixture of gases. A student collects a sample of air and finds that it contains nitrogen, oxygen, carbon dioxide, and trace gases. Which of the following statements is correct?

- (A) Oxygen is the largest component of air.
- (B) Nitrogen is the largest component of air and is mostly inert under normal conditions.
- (C) Carbon dioxide makes up about 21% of air.
- (D) Trace gases like argon and neon do not exist in the air.

29. **Assertion (A):** Polymers like polyester are used to make water-resistant clothing.

Reason (R): Polymers have long-chain molecules that do not absorb water easily.

- (A) Both Assertion and Reason are true, and Reason is the correct explanation of Assertion
- (B) Both Assertion and Reason are true, but Reason is not the correct explanation of Assertion
- (C) Assertion is true, Reason is false
- (D) Assertion is false, Reason is true

30. If you were to observe a "Touch-me-not" (*Mimosa pudica*) plant and a pet dog, which of the following is a life characteristic they BOTH share in a similar way?

- (A) Both can move from one place to another (Locomotion).
- (B) Both respond to the stimulus of touch.
- (C) Both use a mouth to take in nutrition.
- (D) Neither of them produces waste products.

31. Based on the relative positions shown in the diagram, which celestial event is occurring?



- (A) Total Solar Eclipse
- (B) Partial Solar Eclipse
- (C) Total Lunar Eclipse
- (D) New Moon phase

32. Consider the following materials:

- 1. **Diamond**
- 2. **Graphite**
- 3. **Polyethylene**
- 4. **Sodium chloride (NaCl)**

Which of the following correctly pairs the material with its bonding type and characteristic property?

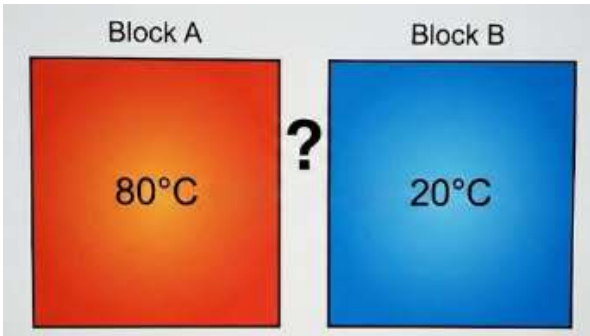
- (A) Diamond → covalent lattice → soft and conducts electricity
- (B) Graphite → covalent layers → conducts electricity
- (C) Polyethylene → ionic → brittle solid
- (D) NaCl → metallic → high melting point

33. A clock with no numbers (only markings) is hanging on a wall. It shows the time as 4:10. What time will it appear to show if seen in a plane mirror opposite to it?

- (A) 4:10
- (B) 8:50
- (C) 7:50
- (D) 10:20

34. A factory emits smoke containing carbon monoxide (CO). "Which of the following explains why people living nearby feel dizzy despite normal oxygen levels?"

- (A) CO reduces the air's oxygen concentration.

- (B) CO binds to hemoglobin, preventing oxygen transport in blood.
 (C) Oxygen in the air is converted to carbon dioxide.
 (D) CO increases nitrogen in the air, reducing oxygen.
35. Which change will make the image in a pinhole camera brighter but blurrier?
 (A) Making pinhole smaller
 (B) Increasing screen distance
 (C) Making pinhole larger
 (D) Removing the screen
36. A chemist has a mixture of fine clay and water. After 1 hour, most clay settles, but the water remains slightly turbid. Which statement is correct?
 (A) Decantation alone will give perfectly clear water.
 (B) Filtration or centrifugation is needed to remove the remaining fine particles.
 (C) Evaporation will remove the clay completely.
 (D) Sublimation can separate clay from water.
37. A student stands 2 m in front of a plane mirror. The mirror is suddenly moved 1 m away from the student. What is the new distance between the student and his image?
 (A) 4 m (B) 5 m
 (C) 6 m (D) 3 m
38. A child uses a kaleidoscope and rotates it slowly. The patterns change because of:
 (A) Absorption of light
 (B) Multiple reflections between mirrors
 (C) Refraction through glass
 (D) Shadow formation light
39. Which material is considered eco-friendly because it is biodegradable?
 A) Plastic B) Glass
 (C) Jute (D) Aluminum
40. On Day 1, the temperature was 20°C. On Day 2, it increased by 10%. On Day 3, it dropped by 2 K. What is the final temperature in Celsius? ($\Delta T_{(K)} = \Delta T_{(C)}$)
 (A) 20°C (B) 18°C
 (C) 22°C (D) 19.8°C
41. Glass is transparent but brittle. Which combination of properties best explains this?
 (A) Amorphous structure and strong intermolecular bonds
 (B) Crystalline structure and weak intermolecular bonds
 (C) Amorphous structure and weak intermolecular bonds
 (D) Crystalline structure and strong intermolecular bonds
42. Two blocks, **Block A** and **Block B**, are kept in contact. An arrow is to be drawn to represent the direction of heat transfer between them. In which direction should the arrow be drawn to show the correct heat flow?
- 
- (A) From B to A
 (B) From A to B
 (C) No heat flows because they are solids.
 (D) Heat flows in both directions equally.

43. Why do some high-quality stainless-steel vessels NOT stick to a magnet, even though they contain iron?
- (A) They are painted with anti-magnetic paint.
- (B) The crystal structure of the alloy (Austenite) changes the magnetic properties of the iron atoms.
- (C) The iron content is less than 1%.
- (D) They are actually made of Aluminum.
44. Which of the following is an "Artificial Satellite" launched by India, primarily used for communication and weather forecasting?
- (A) The Moon (B) INSAT
- (C) PSLV (D) Halley
45. You are looking at the Moon on the 15th day of the lunar cycle (New Moon). Over the next 15 days, the bright part of the moon visible from Earth will:
- (A) Full moon
- (B) Decrease (Waning)
- (C) Remain the same
- (D) Become red
46. A light ray strikes a plane mirror at an angle of incidence of 35° . **What will be the angle between the incident ray and reflected ray?**
- (A) 35° (B) 70°
- (C) 55° (D) 145°
47. Which of the following objects exhibits Rotational, Translatory, AND Periodic motion simultaneously?
- (A) A ball rolling on the ground
- (B) The Earth moving around the Sun
- (C) A drill bit drilling a hole
- (D) A bicycle wheel moving on a road
48. Fish die in a pond during summer even though the water looks clean. Tests show low dissolved oxygen. **Which is the most reasonable explanation?**
- (A) Warm water holds less oxygen, and microbial activity uses up oxygen faster.
- (B) Fish cannot breathe in water with nitrogen.
- (C) Oxygen disappears from water during summer.
- (D) Fish produce carbon dioxide, which replaces oxygen
49. During a science experiment, the room temperature is recorded as 25°C . A liquid is heated until its temperature rises by 45°C . Later, it is allowed to cool, and its temperature drops by 15°C . **What is the final temperature of the liquid on the Kelvin scale?**
- (A) 55 K (B) 328 K
- (C) 318 K (D) 308 K
50. When you blow air through a straw into a test tube containing lime water, the lime water turns milky. This experiment is used to demonstrate **which characteristic of life in humans?**
- (A) Nutrition
- (B) Excretion of Carbon Dioxide during respiration
- (C) Response to stimuli
- (D) Growth

